

Optimizing Drone Delivery Networks Using Artificial Bee Colony Algorithm

Aryan Dhasmana
AIT-CSE
Chandigarh University
Mohali, Punjab
23bcb10081@cuchd.in

Atharva Pratap Singh Chundawat
AIT-CSE
Chandigarh University
Mohali, Punjab
23bcb10067@cuchd.in

Simranjit Kaur
AIT-CSE
Chandigarh University
Mohali, Punjab
23bcb10084@cuchd.in

Vishal Gupta
AIT-CSE
Chandigarh University
Mohali, Punjab
23bcb10080@cuchd.in

Dr. Gurwinder Singh
AIT-CSE
Chandigarh University
Mohali, Punjab, India
gurwinder.e11253@cumail.in

Abstract—This research investigates the Artificial Bee Colony (ABC) algorithm for optimizing drone delivery networks in urban environments, comparing it with traditional path-finding approaches such as Dijkstra's and A-star algorithms. Through experimental implementation across four strategic delivery hubs in Chandigarh, for 110 origin-destination pairs and it has been found that the ABC algorithm achieved superior route optimization in 53.6% of test cases (shorter route length), with an average route distance of 3029.22 meters compared to 3052.22 meters for alternative approaches. However, this modest optimization advantage (approximately, 2530 metres for the chosen pairs) comes with significantly higher computational costs, as ABC required an average execution time of 140.41 seconds (about 2.8 times slower) versus just 36.43 seconds for Dijkstra's algorithm. Also with an optimised drone population the delivery times were reduced by 33.14 %. The study reveals a fundamental trade-off between optimization quality and computational efficiency in drone delivery networks, suggesting that while ABC's ability to balance exploration and exploitation phases makes it well-suited for complex, multi-constraint delivery scenarios, implementation decisions must carefully weigh the benefits of improved routes against increased computational demands, particularly in resource-constrained drone systems with limited onboard processing capabilities.

Keywords—Artificial Bee Colony Algorithm, Path Planning, Drone Delivery Optimization

I. INTRODUCTION

The expansion of drone delivery services is providing new opportunities for logistics, which is expected to reach a \$5.6 billion market by 2030, at a scale of 17.3% per year. This technology offers advantages in delivery speed, operational costs, and impact of logistics on the environment, especially when it comes to last-mile delivery challenges.

Yet, however, drone operations present major challenges including limited flight time (20–30 minutes per fee), regulatory constraints, and the requirement for seamless, challenging urban route planning. These costs make the multi-dimensional optimization problem one that traditional

path-finding algorithms are costly to efficiently resolve. Deterministic algorithms exist such as Dijkstra's and A-star, which provide computational efficiency, but they often do not sufficiently treat the multi-constraint nature of drone delivery optimization.

This paper investigates the implementation of artificial bee colony (ABC) for its bio-inspired swarm intelligence technique—a bio-inspired swarm intelligence technique to optimise drone delivery networks. Through experimental implementation in the urban environment of Chandigarh, we show that the ABC algorithm has better route performance—53.6% optimization over our test cases—than the average route distances of traditional approaches, up to 23 meters.

Our analysis uncovers a core trade-off: the optimization benefits of ABC do however come at a greater computational cost (140.41s vs 36.43s for Dijkstra's). However, the trade-off the algorithm makes between exploration and exploitation phases makes it particularly well suited for these complex, multi-constraint delivery scenarios. Moreover adjusting the drone population for the available demand the overall delivery times were reduced significantly (approx. 30%)

In addition to minimizing costs through route optimization, we have developed First-Come-First-Served (FCFS) based scheduling and scanned for an optimal number of drones per hub depending on the demand.

The results add to the bio-inspired optimization for autonomous systems Aerial logistics, with enormous ramifications for commercial deployment, bases its idea on new drone usage in cities. This paper provides a theoretical background of ABC applied to drone delivery networks and the accompanying performance data of 110 route calculations, revealing the potential for improving the efficiency and scalability of drone delivery networks.

Efficient shortest path algorithms have been developed and have served as the foundation for route optimization research. Gallo and Pallottino [1] gave a thorough investigation of shortest path algorithms and classification based on their methodology and time complexity. Their work laid the foundation for comparing the efficiency of algorithms and compatibility with various genus of networks.

Building on this foundation, Magzhan and Jani [2] noted that assessment of Dijkstra's and other shortest path algorithms—Bellman-Ford and Floyd-Warshall—studies their performance in various network topologies and emphasizes the trade-offs between time complexity and memory use in real-world implementations.

Nature-Inspired Optimization Techniques — has revolutionized methods of route planning. Curkovic and Jerbic [3] pioneered the use of honey-bee behavior to solve problems with path planning, showing its efficacy for navigating complex landscapes featuring obstacles. Saab et al. [4] built off this work and added to the artificial bee colony (ABC) system by better optimizing its capabilities with booster exploitative schemes.

Zhang et al. [5] generalized ABC algorithms to clustering problems, showing the algorithm's versatility beyond path planning and laying the foundations of mathematics and convergence properties that would subsequently be useful in routing applications.

The ABC algorithm has been the subject of attention in recent studies on drone or UAV development and mapping path planning. Aslan and Demirci [6] used ABC for UAV localization issues, showing considerably better accuracy than traditional techniques. An efficient ABC was implemented by Annepu and Rajesh [7] for UAV-assisted wireless sensor node localization using a violet-driven algorithm. These networks are trained to reduce energy usage, thus improving network coverage.

Further advances were made by Muntasha et al. [8], who studied ABC performance for collision avoidance in swarm UAVs. Lin et al. [9] contributed to the development of enhanced multi-strategy ABC algorithms tailored for UAV path planning that combined global exploration with local exploitation.

Saeed et al. [10] showed that swarm intelligence-based strategies may perform better than traditional algorithms for obtaining optimal paths in dynamic environments with multiple drone constraints.

The integration of ABC algorithms with specialized applications has expanded significantly. Çetin Kaya et al. [11] optimized drone routes for COVID-19 medication delivery, while Ghaffar et al. [12] proposed vehicle-UAV integrated routing for emergency medical supplies. Zhang et al. [13] addressed multi-modal transportation through Q-learning under time uncertainty. Alternative approaches emerged alongside ABC, with Huang et al. [14] applying ant colony optimization to vehicle-drone delivery and Cruz et

al. [15] comparing classical shortest path algorithms for in-store navigation. Recent innovations include Chen et al. [16] implementing multi-objective ABC for heterogeneous robotic delivery systems, Elsahy et al. [17] adapting ABC within geographic information systems, and Athanasiadis et al. [18] publishing datasets for van-drone routing problems to facilitate algorithmic comparisons.

III. METHODOLOGY

1.Data sets: Network Construction and Data Preparation

The dataset focuses on the network construction and data preparation for an optimized delivery system in Chandigarh. It includes information derived from a Google Form survey, where customer responses were collected and analyzed to structure an efficient delivery network. As part of this network, four hubs have been strategically mapped onto the map of Chandigarh. Hubs are positioned as Hub A (30.724913, 76.787800), Hub B at (30.746120, 76.769660), Hub C at (30.727379, 76.725188), and Hub D at (30.700463, 76.755896). These locations have been identified to have maximum coverage of the city area. Methods like route optimization, demand forecasting, and predictive modeling can be used. This allows the supply chain to enable businesses to dynamically allocate resources, optimize vehicle dispatching, and mitigate potential delivery bottlenecks.

2.Preprocessing:

The dataset used consists of a Google Form survey, which was preprocessed to ensure correctness and enhance route optimization effectiveness. Data cleaning involved dealing with missing values, deleting duplicates, and applying a process called “geocoding” to convert physical addresses into latitude and longitude coordinates. Outlier detection was conducted using Z-score, IQR filtering, etc., to eliminate extreme values. We then performed data transformation by encoding—one-hot encoding of categorical variables, scaling of numerical data, and dividing up customer locations by their proximity to the four distribution hubs (Hub A, Hub B, Hub C, and Hub D).

Furthermore, distance metrics were calculated for each of the 110 origin-destination pairs, with routes appropriately generated. The final step was to validate the dataset for consistency in a machine-readable structured format, focusing on performance analysis. This refined dataset enabled algorithm testing, runtime measurement, and qualitative assessment integral to effective delivery and Chandigarh network optimization.

3.Experimental Design and Evaluation Matrix Geospatial Visualization

Folium was used to generate our experimental design, creating the delivery network visualization tool with interactive maps and analyzing the best routes predetermined by different algorithms. We initialized four strategic delivery hubs on the map according to their latitude and longitude coordinates. Each hub's service area and optimal routes used heat-map visualizations to show the shortest paths between pairs of origin and destination. We

also mapped out the service areas each hub covers. First, these visualizations illustrate their geographical coverage. For performance analysis, we contrasted the average route distances among traditional algorithms and the Artificial Bee Colony (ABC) algorithm (as one of the optimization algorithms). We also explored the ratio of paths where the shortest path was found by each algorithm, measured with respect to computational load.

To evaluate the computational performance, the total execution time of each algorithm was assessed. The program, executed and plotted over 1000 repetitions, provides the data seen in the program table. In this case, the algorithms were tested with multiple iterations to observe convergence behavior. To assist with this analysis, we also used OSMnx for downloading, modeling, and analyzing street and OpenStreetMap networks, along with NetworkX for graph-based procedures such as routing and community analysis.

To simplify the implementation and to stick to the scope of this paper factors like weather, traffic and other obstructions were kept static. The implementation of drones internal logic was also omitted.

4. Principle of Scheduling in Drone Delivery System

During the early phase of our investigation of the drone delivery system, we developed a scheduling approach focused around the **FCFS (First-Come-First-Serve)** algorithm, concentrating on a single-hub configuration. The FCFS algorithm was chosen for its simplicity and ability to perform tasks in a straight line with little to no variation. This approach ensured that delivery requests were handled precisely in the order they had been received, creating a transparent and equitable system for task ranking and allocation.

To bring this concept to life, we built an extensive dataset that mimicked real-world delivery situations. This dataset included request timestamps and other important parameters such as destination coordinates, package weights, and optional priority levels. We used 100 data points to create a dataset that reflects a wide variety of delivery needs, giving us the ability to stress test the system in different scenarios.

The first step in scheduling was to parse the dataset and use a first-come-first-serve basis to assign tasks to available drones upon request arrival, while also optimizing the flight trajectories to minimize lead times and energy usage. By starting with the FCFS algorithm and a single-hub setup, we aimed to identify the foundational challenges specific to drone scheduling—such as battery health optimization, resource allocation, and collision avoidance.

This initial configuration not only served as infrastructure for system evaluation but also provided valuable insights into understanding the behavior of a drone delivery system under controlled conditions.

These results will pave the way toward future investigations into advanced scheduling techniques. In the next phases of our research, we plan to explore route policies such as **priority-based** or **dynamic scheduling**.

This step-by-step process results in a comprehensive understanding of what the system can and cannot do—laying the groundwork for more advanced and smarter drone delivery solutions.

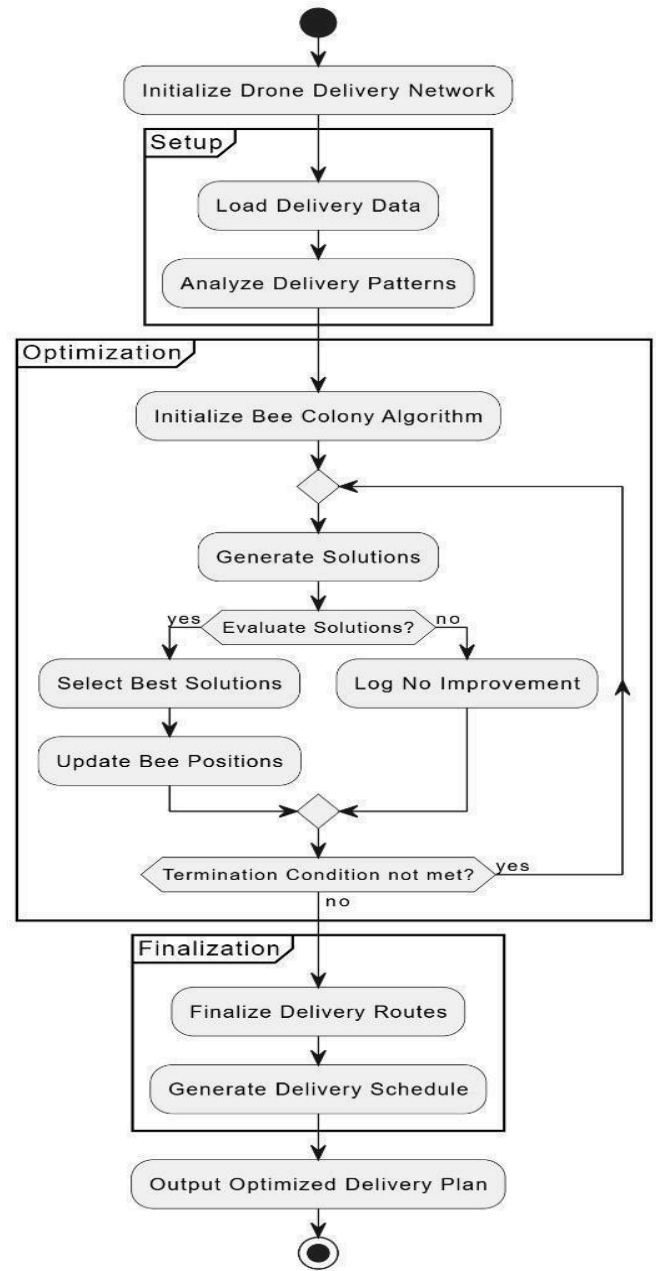


Fig.1 : System Flow

5. Mathematical Interpretation of Algorithms

5.1 Dijkstra's Algorithm

Finding the shortest path between two nodes in a weighted graph may be done using Dijkstra's Algorithm.

5.1.1 Graph Representation

Let $G = (V, E)$ be a weighted graph, where V is the set of vertices and E is the set of edges. Each edge $(u, v) \in E$ has a weight $w(u, v)$, represents the cost/ distance between nodes.

5.1.2 Mathematical Formulation

1. Initialize the distance to all vertices as infinity :

$$d(v) = \infty, \forall v \in V$$
2. Set the source vertex s with $d(s) = 0$.
3. Update the distance of each vertex v using the relaxation function :

$$d(v) = \min(d(v), d(u) + w(u, v)), \forall (u, v) \in E$$
4. Use a priority queue always to expand the vertex with the smallest distance

5.1.3 Time Complexity

$O((V + E) \log \log V)$ using the priority queue.

5.2 Artificial Bee Colony (ABC) Algorithm :

ABC is an optimization algorithm which is inspired by nature, specifically from the foraging behaviour of honeybees.

Here, the number of bees(scouts) was kept constant at 50 and the iterations varied from 100 to 1000.

5.2.1 Mathematical Representation:

Initialization Phase

(a) **Population Initialization** : Let S be the set of all possible solutions (hubs). The initial population P of n bees is generated :

$$P = \{s_1, s_2, \dots, s_n\} \text{ where } s_i \in S$$

Each bee i is assigned a random solution s_i .

(b) **Fitness Evaluation** : The fitness of each solution s_i is calculated :

$$f(s_i) = d(s_i, u)$$

where $d(s_i, u)$ is the distance from the hub s_i to the user's location u .

Employed Bees Phase

Local Search: For each bee i :

- A new solution s'_i is randomly chosen from S .
- The fitness of the new solution is evaluated $f(s'_i) = d(s'_i, u)$
- If the new solution is better (has a lower distance) :

$$\text{if } f(s'_i) < f(s_i) \text{ then } s_i \leftarrow s'_i$$

Onlooker Bee Phase

(a) Probabilistic Selection: The probability $p(s_i)$ of selecting a solution s_i for further exploration is calculated :

$$p(s_i) = \frac{\frac{1}{f(s_i)}}{\sum_{i=0}^n \frac{1}{f(s_i)}}$$

Where $\sum_{i=0}^n \frac{1}{f(s_i)}$ is the sum of the inverse fitness values for all solutions.

(b) Selection and Local Search: For each onlooker bee:

- A solution s_i is selected based on the probability $p(s_i)$.
- A new solution s'_i is randomly chosen from S .
- The fitness of the new solution is evaluated as :

$$f(s'_i) = d(s'_i, u)$$
- If the new solution is better:

$$f(s'_i) < f(s_i) \text{ then } s_i \leftarrow s'_i$$

Scout Bees Phase

Replacement: If a solution s_i has not been improved after a certain number of iterations (or it's fitness is the worst in the population)

- Replace s_i with the new random solution s'_i :

$$s_i \leftarrow s'_i \text{ where } s'_i \in S$$

- Evaluate the fitness of the new solution :

$$f(s'_i) = d(s'_i, u)$$

Optimization Goal

The overall goal of the ABC algorithm is to minimize the fitness function:

$$f(s)$$

Where S is the set of all possible solutions (hubs).

Termination

The algorithm terminates after a predefined number of iterations

5.2.2 Time Complexity: $O(N \cdot M \cdot I)$

Where:

- N is the number of food sources,
- M is the number of iterations, and
- I is the number of bees.

5.2.3 Abstract:

The Artificial Bee Colony (ABC) algorithm is an optimization method inspired by honey bee foraging behavior. It provides a resilient and adaptable approach to solving optimization problems. By leveraging the roles of **employed**, **onlooker**, and **scout** bees, ABC effectively balances exploration and exploitation—making it suitable for a wide range of applications.

Variants like **CNSABC** have been developed to better serve specific optimization needs. The ABC algorithm has maintained over **twenty years of relevance and successful use** in the field of swarm intelligence optimization.

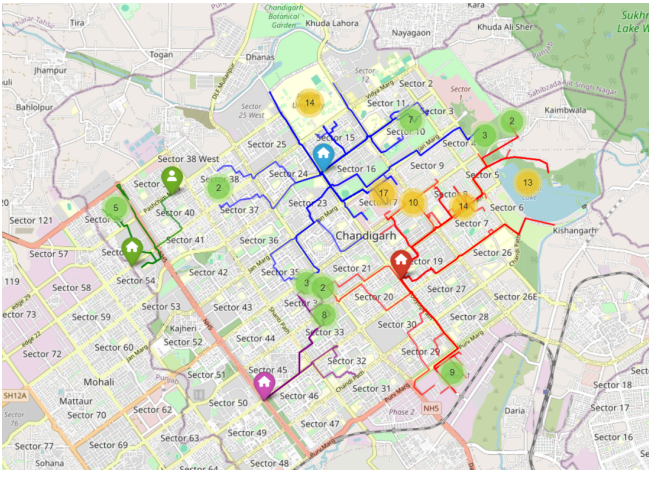


Fig.2 : Paths traced using ABC Algorithm

5.3 A-Star Algorithm

A* (A-star) is a complex and advanced algorithm methodology for **pathfinding** and **graph traversal**, created to hierarchically determine the shortest path between nodes in a graph more efficiently. It incorporates the principles of **Dijkstra's algorithm**, which computes the minimum distance from a starting node, along with the **heuristic-guided** approach of **Greedy Best-First Search (GBFS)**, which prioritizes nodes based on their estimated distance to the goal.

5.3.1 Heuristic Function

Uses a heuristic $h(u)$ to estimate the cost from node v to the goal.

Total cost Function :

$$f(v) = g(v) + h(v)$$

Where:

- $g(v)$: The actual cost from the start node to v .
- $h(v)$: The estimated cost from v to the goal.

5.3.2 Mathematical Formulation

Initialize:

$$g(s) = 0, \quad f(s) = h(s).$$

Expand the node with the lowest $f(v)$.

Update costs:

$$g(v) = (g(v), g(u) + w(u, v))$$

$$f(v) = g(v) + h(v)$$

Stop when the goal is achieved.

5.3.3 Time Complexity

$O((V + E) \log V)$ in the worst case.

IV. FINDINGS

The findings presented strongly highlight the advantages of the Artificial Bee Colony (ABC) algorithm in optimizing delivery networks. The comparative analysis between the ABC algorithm and Dijkstra's algorithm demonstrates that ABC achieves slightly better efficiency in terms of delivery times and travel distances, which are critical for drone delivery optimization.

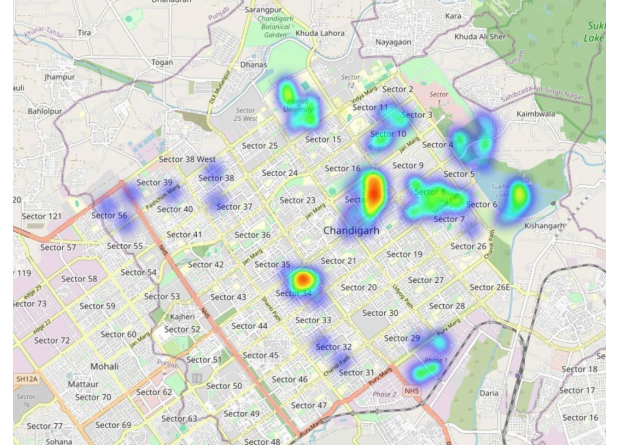


Fig.3 : Heatmap showcasing data spread

The performance of ABC algorithm varies with the number of iterations. ABC algorithm, as iterations increase from 100 to 1000, the average time increases by 144.7%, from 57.35 to 140.32 seconds. Conversely, the total distance decreases by 22.8% over the same range, from 431.68 to 333.21. The most significant improvement in total distance (11.7%) occurs between 100 and 200 iterations. After 500 iterations, the improvement in total distance is minimal, with only a 1.9% decrease from 500 to 1000 iterations. The average time increases more rapidly after 500 iterations, with a 67.1% jump from 500 to 1000 iterations. Overall, the best balance between time and distance improvement appears to be around 300-500 iterations.

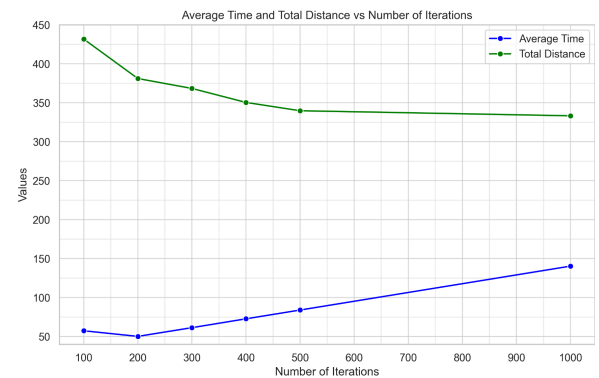


Fig.4 : ABC relation between iterations and compute time

In terms of delivery time, the ABC algorithm achieves an average total delivery time of 9.19 minutes compared to Dijkstra's 9.24 minutes, with a maximum total delivery time of 17.75 minutes for ABC versus 17.21 minutes for Dijkstra.

The difference in total minutes is slightly higher for ABC (8.56 minutes) compared to Dijkstra (7.97 minutes), suggesting a marginal improvement in scheduling efficiency. Similarly, travel time comparisons show that ABC has an average travel time of 4.54 minutes and a maximum travel time of 8.57 minutes, closely matching Dijkstra's average travel time of 4.58 minutes and maximum travel time of 8.60 minutes, with both algorithms exhibiting identical differences in travel times (4.03 minutes). The graph comparing travel minutes further illustrates that both algorithms perform similarly across test scenarios, with overlapping trends in travel times.

TABLE I. PERFORMANCE METRICS

Name	Distances		
	Total distance (metres)	Mean(km)	Median(km)
Dijkstra	335744.383	3052.221	2864.575
ABC	333214.708	3029.224	2860.305
A*	335744.383	3052.221	2864.575

Route optimization is a key strength of the ABC algorithm, as it achieves a shorter mean distance of 3029.22 meters compared to Dijkstra's 3052.22 meters, representing a modest but consistent improvement in delivery efficiency. This aligns with the goal of minimizing travel distances in drone delivery networks and highlights the ABC algorithm's ability to optimize routes effectively.

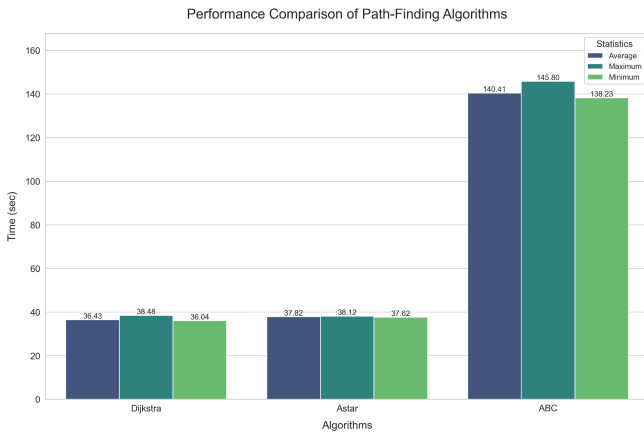


Fig.4 : Time Analysis

However, computational performance reveals a significant trade-off for using the ABC algorithm. While Dijkstra operates at an average execution time of just 36 milliseconds, ABC incurs substantially higher computational costs, averaging over 140 seconds per execution. Other algorithms like A-star and Bellman-Ford(BLF) also outperform ABC in terms of computational efficiency, requiring only about 37 seconds and 115 seconds, respectively. Despite this drawback, the improved route optimization provided by ABC makes it valuable for applications where delivery efficiency is prioritized over computational speed.

TABLE II. HUB CONFIGURATION

Algorithm	Hub Drone Configurations		
	Best(min)	Worst(min)	Percentage Change
ABC	499.82	747.61	33.14 %
Dijkstra	503.62	729.03	30.92 %
A*	503.62	729.03	30.92%

The ABC Algorithm's best configuration, Hub:Drones{A:3, B:3, C:1, D:2} achieved a Total Delivery Time of 499.82 minutes, while its worst configuration (1 each) resulted in 747.61 minutes. In contrast, the Dijkstra and A* Algorithm's best configuration {A:3, B:2, C:1, D:2} yielded a delivery time of 503.62 minutes, with the same worst configuration leading to 729.03 minutes. This provides us with more than a 30% decrease in delivery times.

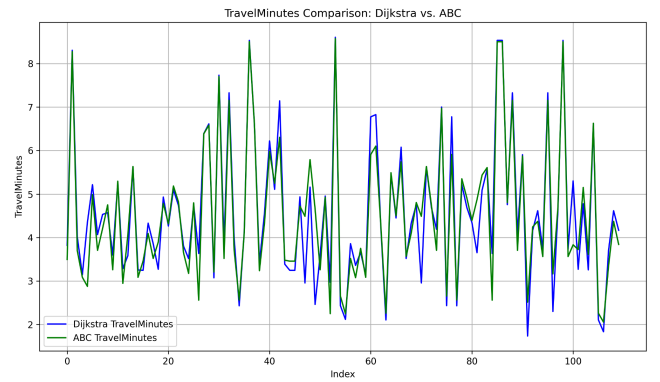


Fig.5: Time Analysis

All in all, the ABC algorithm proved to be able to save total delivery time and travel distances given that vehicle utilization, in general, should not be too high. The performance of the technique achieved reflects its higher computational cost, which might make it not readily applicable in resource-constrained environments, but the efficiency in optimizing routes and maximizing scheduling makes this technique the best candidate for enhanced logistics in aerial systems, where the performance of the system is critical.

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